

QR CODE TOOLS MANAGEMENT SYSTEM

Complete User Guide

& User Flow Reference

Admin

Inspector

Store Manager

Worker

Management

Data Entry

Construction & Industrial Equipment Tracking

Version 1.0 | May 2026 | For Internal Use Only

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1 System Overview

The QR Code Tools Management System is a digital platform for tracking construction and industrial tools across multiple job sites. It replaces manual paper logs with a smart, scannable, role-based system that keeps every stakeholder informed in real time.

Core Capabilities

Capability	Description
Track Tools	Know where every tool is, its condition, and who last used it.
Scan QR Codes	Instantly pull up any tool's details by scanning its QR sticker.
Run Inspections	Record safety checks with photos and usability ratings.
Move Tools	Log when tools move between sites and subcontractors.
Get Alerts	Automatic warnings for expired or damaged tools.
Generate Reports	Export tool lists and certificates as Excel or PDF.
Match Split Tools	Verify that Part A and Part B of a tool belong together.
Bulk Import	Upload Excel files to register many tools at once.

2 User Roles & Access Control

Each user is assigned a role that controls which pages they can access and what actions they can perform. After login, users are automatically redirected to their role-specific page.

Role	Pages Accessible	Can Edit?
Admin	All pages	Yes — full control
Data Entry	Tool Master (own site only)	Yes — create tools only
Inspector	Inspector View	Yes — inspections only
Store Manager	Store View	Yes — transactions only
Worker	Worker View	No — read-only
Management	Dashboard, Reports, Alerts	No — read-only

IMPORTANT

Contact your system administrator if you need access to pages not listed for your role.

3

Getting Started — Login Flow

Open your browser and navigate to the system URL provided by your administrator (e.g., `http://192.168.x.x:5173`). You will be greeted by the login screen.

User Flow — Login Process

Step 1: Open browser → Navigate to system URL

Step 2: Login screen appears

Step 3: Enter Username and Password

Step 4: Click [Sign In]

■ Valid credentials? ■

■ YES → JWT token issued → Redirect to role ■

■ NO → Error message shown → Retry ■

Alternative: Scan your personal QR code to authenticate

TIP

If you cannot log in, contact your admin to verify your account is Active.

4 Admin Guide — Tool Master

The Tool Master is the central hub for creating, viewing, and editing all tool records. It contains three tabs: New Entry, Saved Tools, and Import.

4.1 Creating a New Tool

- Step 1: Click 'Tool Master' in the sidebar
- Step 2: Select the 'New Entry' tab
- Step 3: Fill in the form (see field table below)
- Step 4: Upload Test Certificate (PDF/JPG/PNG) – optional
- Step 5: Click [Save Tool]
- Step 6: System auto-generates a unique QR code
- Step 7: 'New Tool Added' info alert is created automatically
- Step 8: QR code displayed – print or download

Field	What to Enter	Required?
Description	Name of the tool (e.g., Clamp Head)	YES
Make / Year	Manufacturer or year of manufacture	YES
Capacity	Load capacity (e.g., 8 Ton)	YES
Safe Working Load	Maximum safe load rating	YES
Tool Type	Erection Tools or Stringing Tools	YES
Metal Type	Material the tool is made from	YES
Tool Variant	Specific variant or model	YES
Purchaser Name	Who purchased the tool	YES
Purchaser Contact	Purchaser phone number	No
Supplier Code	Supplier reference number	No
Date of Supply	When the tool was received (YYYY-MM-DD)	YES
Validity Period	How many years the tool is certified for	YES
Test Certificate	Upload PDF or image file	No
Current Site	Where the tool is located now	YES
Subcontractor Name	Name of the subcontractor (if applicable)	No
Subcontractor Code	Subcontractor reference code	No
Subcontractor Mobile	Subcontractor phone number	No
Job Code	Project reference code	No
Job Description	Project description	No

Field	What to Enter	Required?
Remarks	Any additional notes	No

QR Code Auto-Generation Logic

The system builds each QR code automatically from the tool's data:

Name Part : First 2 letters of description (e.g., 'CL' for Clamp)
Metal Part : First letter of metal type (e.g., 'S' for Steel)
Variant Part : Initials of each word in variant (e.g., 'TA' for Type A)
Capacity Part: Numeric digits from capacity (e.g., '8' from '8 Ton')
Date Part : MM-YY from date of supply (e.g., '01-24')
Supplier Part: First 2 letters of purchaser (e.g., 'AB' for ABC Corp)
Serial Part : 4-digit auto-incremented number (e.g., '0101')

Example result: CLSTA8-01-24AB0101

Uniqueness guaranteed – system appends '-1', '-2' if duplicate found.

4.2 Viewing & Editing Tools

- Step 1: Go to Tool Master → 'Saved Tools' tab
- Step 2: Use the search bar to find by any field
OR use the Site / Creator filter dropdowns
- Step 3: Click a tool row to view full details
- Step 4: Click the Edit icon to modify the tool
- Step 5: Make your changes and click Save
NOTE: If the site field changes, a movement history entry
is recorded automatically
- Step 6: Confirmation message displayed

4.3 Printing QR Codes

- Step 1: Go to Tool Master → Saved Tools tab
- Step 2: Find the tool in the table
- Step 3: Click the QR icon on that row
- Step 4: QR code image appears in a modal popup
- Step 5: Click Print or Save Image
- Step 6: Print the label and stick it on the physical tool

5 Admin Guide — Dashboard

The Dashboard provides a real-time snapshot of your entire tool inventory through summary cards and interactive charts.

Summary Metric Cards

Card	Meaning
Total Tools	All tools registered in the system
Usable	Tools currently in good working condition
Scrap	Tools retired after a failed inspection
Expiring Soon	Tools whose certification expires within 30 days
Overdue	Tools whose certification has already expired

Available Charts

Chart	Type	What It Shows
Tools by Site	Bar	How many tools are at each location
Status Distribution	Pie	Usable vs. Scrap split
Inspection Trends	Line	Number of inspections over time
Tool Age Distribution	Bar	Breakdown of tool ages
At-Risk Tools	List/Bar	Tools with low usability or near expiry

User Flow — Using the Dashboard

- Step 1: Click 'Dashboard' in the sidebar
- Step 2: Review the five summary cards at the top
- Step 3: (Optional) Select a filter: by User or by Site
- Step 4: Review charts for trends and problem areas
- Step 5: Click 'Export' to download a data snapshot

6 Admin Guide — Alerts

The Alerts system automatically monitors all tools and notifies administrators when action is required. Alerts are sorted by severity — most critical first.

Alert Types & Severity

Alert Level	Colour	Trigger	Action Required
	Red	Expired tool / Usability <80% / Scrap	Immediate
Warning	Yellow	Expiring within 30 days	Within 30 days
	Blue	New tool / New user / Pass inspection	None

When Are Alerts Triggered?

Alert	Triggered When
New Tool Added	A tool record is created in the system
New User Created	A user account is added
Tool Expiring Soon	Expiry date is within 30 days of today
Tool Expired	Expiry date has already passed
Low Usability	Inspection usability result is below 80%
Tool Scrapped	A tool is marked as Scrap after inspection

User Flow — Managing Alerts

- Step 1: Click 'Alerts' in the sidebar
- Step 2: Alerts displayed grouped by severity (Critical first)
- Step 3: Review all RED (Critical) alerts first
- Step 4: Click any alert to see full details:
 - Tool ID and description
 - Site location
 - Timestamp of trigger
- Step 5: Take appropriate action:
 - Critical → Withdraw tool / Schedule immediate inspection
 - Warning → Plan inspection or replacement within 30 days
 - Info → No action needed

SAFETY NOTE

Critical alerts indicate tools that pose a safety risk. Never ignore red alerts.

7 Admin Guide — Users Management

Admins manage all user accounts from the Users page. You can add, edit, deactivate, or delete users.

User Flow — Add a New User

- Step 1: Click 'Users' in the sidebar
- Step 2: Click the [Add User] button
- Step 3: Fill in:
 - Full Name
 - Username (unique login name)
 - Email address (validated format)
 - Phone number
 - Password
 - Role (Admin/Inspector/Store/Worker/Management/Data Entry)
 - Assigned Site
- Step 4: Click Save
- Step 5: System creates the account and triggers a 'New User' info alert
- Step 6: User appears in the list and can now log in

User Flow — Deactivate a User

- Step 1: Find the user in the list
- Step 2: Toggle the Active/Inactive switch to INACTIVE
- Step 3: User immediately loses login access
 - NOTE: Account and all history are preserved (not deleted)
- Step 4: Toggle back to ACTIVE to restore access

User Flow — Delete a User

- Step 1: Find the user in the list
- Step 2: Click the Delete icon
- Step 3: Confirm the deletion in the dialog
- Step 4: User is permanently removed (Admin only)
 - WARNING: This action cannot be undone

RECOMMENDATION

Prefer Deactivating users over Deleting them to preserve audit history.

8 Admin Guide — Reports

The Reports page lets you search across all tool data and export the results as a CSV file compatible with Excel or Google Sheets.

User Flow — Exporting a Report

- Step 1: Click 'Reports' in the sidebar
- Step 2: Type in the search box to filter by any field
(description, QR code, site, subcontractor, etc.)
- Step 3: Use column filter dropdowns to narrow results further
- Step 4: Hover over any row to see the QR code preview
- Step 5: Click [Download CSV]
- Step 6: CSV file saved to your computer
(Opens in Excel, Google Sheets, or any spreadsheet app)

9 Inspector User Guide

Inspectors are automatically taken to the Inspector View after login. This is the only page available to this role. Your job is to find tools and record their safety inspection results.

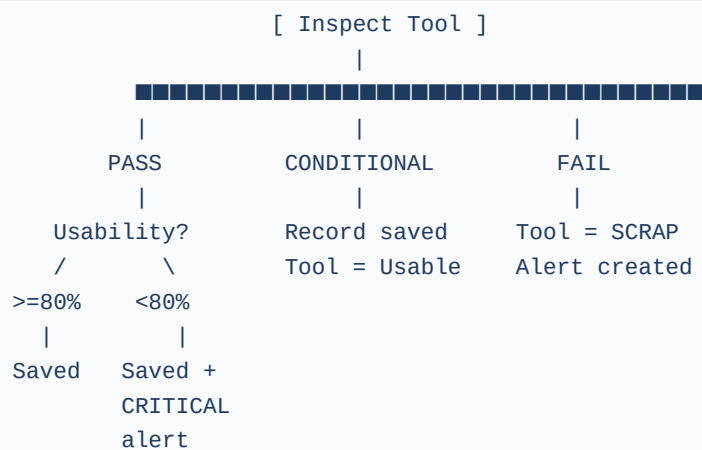
User Flow — Complete a Safety Inspection

- Step 1: Inspector View opens automatically after login
- Step 2: Find the tool using ONE of these methods:
- a) Camera Scan – Click camera icon, point at QR sticker
 - b) Upload Image – Photograph the QR sticker, upload the photo
 - c) Manual Search– Type the QR code or tool ID in search box
- Step 3: System fetches and displays tool details:
- Description, Make, Capacity, SWL
 - Current site, Status, Last inspection date
- Step 4: Physically review the tool
- Step 5: Fill in the Inspection Form:
- Result : Pass / Conditional / Fail
 - Usability % : Drag slider 0-100
 - Photos : Upload damage/condition photos
 - Remarks : Write your observations
- Step 6: Click [Submit Inspection]

What Happens Automatically After Submission

Condition	Automatic Action
Any result	last_inspection_date updated to today
Any result	usability_percentage updated to entered value
Result = Fail	tool.status changed to 'Scrap' immediately
Usability < 80%	Critical alert created for admin
Result = Pass	Confirmation message shown — no further action

Inspection Decision Flow



IMPORTANT

IMPORTANT
A Fail result permanently marks the tool as Scrap. Double-check before selecting Fail.

10 Store Manager User Guide

Store Managers are automatically taken to the Store View after login. Your job is to record all tool movements into and out of your store location.

User Flow — Record a Tool Transaction

- Step 1: Store View opens automatically after login
- Step 2: Scan the tool QR code using camera
OR type the QR code manually in the search box
- Step 3: System displays tool details and last transaction
- Step 4: Select Transaction Direction: IN or OUT
- Step 5: Select Transaction Type (see table below)
- Step 6: Fill in additional details:
 - Subcontractor Name, Code, Mobile (if applicable)
 - Target Site (where the tool is going)
 - Remarks (optional)
- Step 7: Click [Record Movement]
- Step 8: System updates tool's current site and records history
- Step 9: Last transaction confirmation displayed on screen

Transaction Types

Direction	Type	When to Use
IN	subcon_return	Subcontractor returning a tool
IN	new_product	Brand new tool arriving for first time
IN	site_receive	Tool transferred from another site
OUT	subcon_work	Sending tool to a subcontractor for work
OUT	site_transfer	Transferring tool to another site

User Flow — Incident Mode

Use Incident Mode when recording a damaged or missing tool.

- Step 1: Toggle 'Incident Mode' ON before scanning the tool
- Step 2: Scan the tool as normal
- Step 3: Enter the 'Debit To' code (person/team responsible)
- Step 4: Record movement as normal
- Step 5: An alert is created for the admin automatically

NOTE

Every movement is logged with your name, the tool, and a timestamp for full audit trail.

11 Worker User Guide

Workers are automatically taken to the Worker View after login. This is a read-only view — no changes can be made. Use it to check a tool's status and safety information before use.

User Flow — Check a Tool

- Step 1: Worker View opens automatically after login
- Step 2: Click the [Scan] button
- Step 3: Point your camera at the tool's QR sticker
- Step 4: System displays all tool information:
 - Description & Make
 - Capacity & Safe Working Load (SWL)
 - Current Site
 - Status: USABLE (green) or SCRAP (red)
 - Usability %
 - Expiry Date
 - Subcontractor Info

Worker Decision Guide

Status	Usability	Decision
USABLE	$\geq 80\%$	Safe to use
USABLE	$< 80\%$	Use with caution — report to supervisor
SCRAP	Any	DO NOT USE — Report to supervisor immediately
USABLE (Expired)	Any	DO NOT USE — Certification expired

SAFETY RULE

If the tool shows SCRAP or an expired date — stop and report it to your supervisor immediately.

12 Split Tool Matching

Some tools consist of two parts (Part A + Part B) that must be used together. The Split Tool Matching feature verifies that you have the correct matching pair before use.

User Flow — Match a Split Tool

- Step 1: Navigate to the Split Tool page
- Step 2: Scan Part A:
- Click [Scan Part A]
 - Scan or upload the QR code from the first component
 - Part A details displayed on screen
- Step 3: Scan Part B:
- Click [Scan Part B]
 - Scan or upload the QR code from the second component
 - Part B details displayed on screen
- Step 4: System automatically compares:
- ✓ Description match?
 - ✓ Current site match?
 - ✓ Make match?
 - ✓ Capacity & SWL match?
- Step 5: Result displayed with colour indicator

Matching Results

Result	Indicator	Meaning	Action
Correct Combination	GREEN	Parts match, both Usable	Safe to use
Mixed Status	YELLOW	Parts match, one/both not fully usable	Caution — report to supervisor
Mismatch	RED	Description or site don't match	STOP — Wrong combination

SAFETY RULE

Never use a RED (Mismatch) combination. Always confirm GREEN before use.

13 Bulk Import Guide

Use bulk import to register many tools at once by uploading an Excel file, instead of entering them one by one. The system will automatically generate QR codes for every imported tool.

User Flow — Bulk Import Tools

- Step 1: Go to Tool Master → 'Import' tab
- Step 2: Click [Download Template] to get the Excel template file
- Step 3: Open the file in Excel or Google Sheets
- Step 4: Enter one tool per row (see column guide below)
- Step 5: Save the file
- Step 6: Back in the system – click [Upload File]
- Step 7: Select your completed Excel or PDF file
- Step 8: Click [Import]
- Step 9: System processes each row:
 - Validates required fields
 - Generates a unique QR code per tool
 - Creates each Tool record in the database
- Step 10: Download your results:
 - QR Code ZIP : PNG images of all QR codes, ready to print
 - Excel + QR : Original file with 'qr_link' column added
- Step 11: Any failed rows are listed with the reason – fix and re-upload

Excel Column Reference

Column Name	Required?	Example Value	Notes
description	YES	Clamp Head	Name of the tool
make	YES	2022	Year or manufacturer
capacity	YES	8 Ton	Load capacity
safe_working_load	YES	8 Ton	Maximum safe load
purchaser_name	YES	ABC Corp	Who purchased the tool
supplier_code	No	SUP001	Supplier reference
date_of_supply	No	2024-01-15	Format: YYYY-MM-DD
tool_type	No	Erection Tools	Erection or Stringing
metal_type	No	Steel	Material
tool_variant	No	Type A	Specific model/variant
purchaser_contact	No	+1234567890	Phone number
job_code	No	JOB-001	Project reference
job_description	No	Tower Erection	Project description

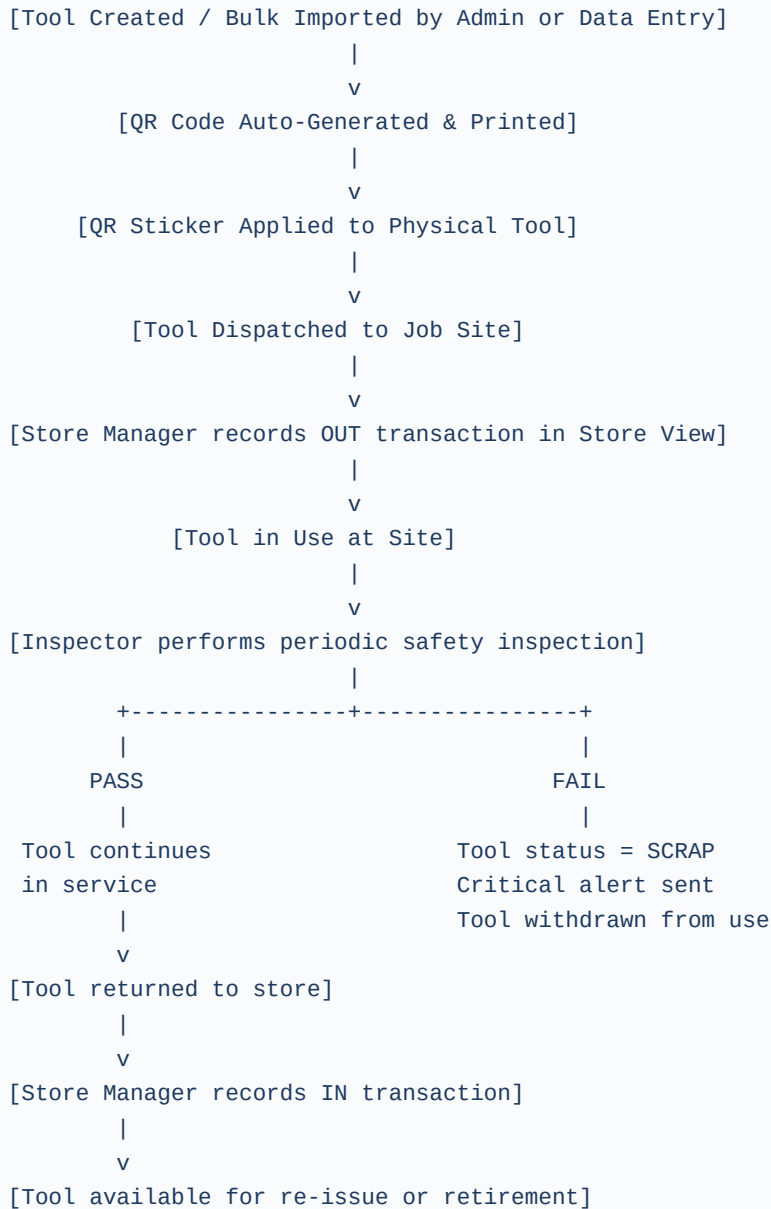
Column Name	Required?	Example Value	Notes
location	No	Site A	Current site name

Common Import Errors

Error	Cause	Fix
Missing required field	Column left blank for a row	Fill in the missing value
Invalid date format	Date not in YYYY-MM-DD format	Change to e.g. 2024-01-15
Duplicate entry	Tool already exists in system	Remove the duplicate row

14 Complete System Flow Diagram

Overall Tool Lifecycle Flow



Alert Monitoring Flow (Continuous)

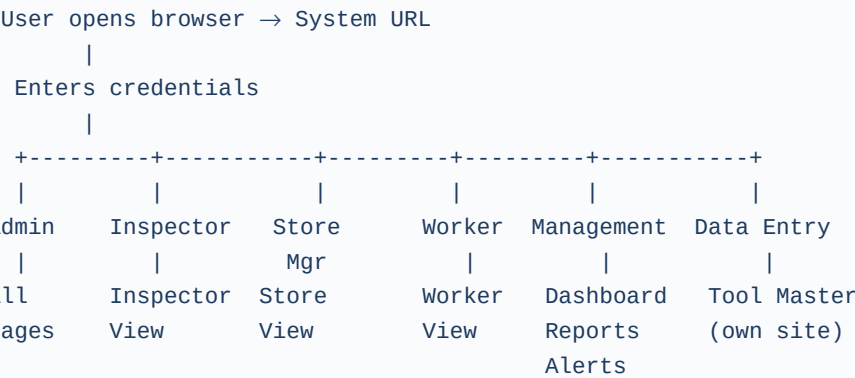
System continuously monitors all tools:

- Expiry date within 30 days? → WARNING alert created
- Expiry date has passed? → CRITICAL alert created
- Usability < 80%? → CRITICAL alert created
- Tool status = Scrap? → CRITICAL alert created
- New tool added? → INFO alert created
- New user added? → INFO alert created

Admin reviews alerts daily:

- CRITICAL → Immediate action – withdraw / replace tool
- WARNING → Schedule action within 30 days
- INFO → No action required

Login & Role Routing Flow



15 Frequently Asked Questions

Q1: I can't log in. What do I do?

A: Contact your admin to verify your username and password are correct, and that your account is set to Active status.

Q2: The QR scan isn't working. What should I try?

A: (1) Make sure your browser has camera permission. (2) Try the 'Upload Image' option — photograph the QR sticker and upload the photo. (3) If the code is damaged or too small, type it manually in the search box.

Q3: A tool shows as Scrap but it's still usable. Can I change it?

A: Only an Admin can update a tool's status. Contact your administrator to review and correct the record after verifying the tool's actual condition.

Q4: I made a mistake when adding a tool. Can I fix it?

A: Yes. Go to Tool Master → Saved Tools, find the tool, click the Edit icon, make your corrections, and save.

Q5: The wrong QR code is on a physical tool. What do I do?

A: Find the correct tool record in Saved Tools. Print the correct QR code from that record. Peel off the wrong sticker and replace it with the printed correct one.

Q6: How do I know if two tool parts belong together?

A: Use the Split Tool Matching feature. Scan both Part A and Part B. The system tells you instantly if they are a correct combination (green), mixed status (yellow), or a mismatch (red).

Q7: My bulk import has errors. What do I do?

A: After import, the system lists every row that failed and explains why. Fix those rows in your Excel file and re-upload only the failed rows.

Q8: Can I use this system on my phone?

A: Yes. Open the system URL in Chrome or Safari on your phone. The QR scanning feature uses your phone's camera directly.

Q9: How long are tool records kept?

A: Tool records, inspection history, and movement logs are kept indefinitely. Admins can export a PDF report covering the last 12 months of activity.

Q10: What happens when I submit a Fail inspection?

A: The tool is automatically marked as Scrap and a Critical alert is sent to the admin. The tool must be withdrawn from service immediately. This action cannot be undone by an inspector — contact an admin if a correction is needed.

Q11: Who can see which tools?

A: Admins and Management can see all tools across all sites. Data Entry users can only see tools at their assigned site. Inspectors, Store Managers, and Workers can look up any tool by scanning its QR code.

Q12: How do I export a tool's certificate?

A: Go to Reports, find the tool, and click its row. You can view and print the tool certificate as a PDF from the tool's detail view.

For technical support or to report a system issue, contact your system administrator.

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